Ability score allocation systems in RPGs

a treatment of the D&D point-buy system and generalizations

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In this work we examine D&D5's ability score allocation variant (point-buy) system, how it works and what scorre palettes it yields as a result. We then generalize to other possible compatible systems to extract possible features a good score allocation system has. Finally after proposing natural extentions of D&D5's point-buy systems, we set down the rules for creating any (parametrized) custom system from scratch.

1. D&D5's standard

1.1. D&D5's abilities

Most RPGs nowadays follow an ability score system, by which a set of distinct abilities is assigned scores to quantify a character's said abilities. In this work we will follow the example of D&D5's score system as a template.

In D&D a character is parametrized by 6 abilities as shown in Table 1.1, for which the range is 1 to 20, so 1 is the lowest possible value instead of 0, and the total number of possible values are 20. This is an aesthetic choice, and it simply stands for not allowing a character to embark on an adventure without an ability baseline.

	ability	measuring
STR	strength	physical power
DEX	dexterity	agility
CON	constitution	endurance
INT	intelligence	reasoning and memory
WIS	wisdom	perception and insight
CHA	charisma	force of personality

Table 1.1: D&D abilities

At this point it is prudent to mention that the effect of said scores for each ability are represented by an action modifier that is defined by

$$\text{modifier} = \frac{\text{score} - \dfrac{\max \text{score}}{2}}{2}$$

which results in the modifiers listed in Table 1.2. This will justify the use of a statistic in evaluating a score palette later.

score	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
modifier	-5	-4	-4	-3	-3	-2	-2	-1	-1	0	0	+1	+1	+2	+2	+3	+3	+4	+4	+5

Table 1.2: D&D modifiers

1.2. D&D5's point-buy

The D&D standard states that you can allocate ability "point" from a "pool of points" but in fact there is more to that. As can be seen in Table 1.3 however, ther eis a complication: suddently for going from 13 to 14 requires 2 points spent, so it is not exactly an ability allocation system per se, but one that relies on an "artificial" currency by which ability points may cost 1 or may cost more if score goes too high. This is common in other RPGs like the computer game "Knights of the Old Republic" which makes higher 1 point increments even more costly.

score	8	9	10	11	12	13	14	15
cost of acqisition	0	1	2	3	4	5	7	9
cost of increment		+1	+1	+1	+1	+1	+2	+2

Table 1.3: D&D point-buy costs

This nonlinearity in costing ability scores allocation introduces some very interesting game balance aspects to a game. You stand to loose more than you normally would for every "high" score you pick, there is a "loss" in the exchange which is different to a balanced trade-off (without losses). Imagine as an analogy having three coins and tossing two of them to your friend, but one drops on the ground so they only get one. If the trade-off wasn't lossy, you should loose two coins and your friend should get 2 coins. The +2s in the cost "schema" indicate that getting these high scores is a lossy trade-off in terms of ability points: for every high score point you add, you loose two low ability points, not one, due to the higher price of ability points added to high scores.

1.2.1. D&D5's schema basic features

Lets examine D&D's schema in Table 1.3 on the previous page mroe closely; it has some minimal identifiable features:

baseline To identify a minimum score, it is not sufficient to limit the range of allowable scores; the lowest allowable score must also (by construction) have cost of acquisition 0 (while cost of increment is not defined for that). While it is not mathematically illegal to have a non-zero cost for baseline, it really does not make sense, as a baseline represents that these base scores are "given", so "free of cost".

positive costs of acquisition Axiomatically, only positive scores of acquisition make sense, apart from the baseline, by definition.

positive costs of increment (strictly-increasing cost of acquisition) The least one can do is increase the cost +1 for each ability point increase, which leads to a trivial straightforward ability point allocation system. Having a cost 0 cost of increments makes a particular score "pass-through", effectively deleting it from the schema.[1] This means that costs of increment are in fact strictly increasing.

Some games prefer to display their schema with cost increments instead, to better show how the costs ramp up. D&D prefers to show cost of acquisition instead. Whichever format is used to add ability points they all eventually answer to a cutoff sum that limits players in creating characters; for example D&D players cannot exceed the cost of 27.

1.2.2. D&D5's schema advanced features

Apart from the basic features that make a schema functional in a game, there are some finer details that actually make a particular schema more game-balanced:

increasing cost of increment The lowest bound of +1 does not normally prevent highs and lows in costs of increments; for example a $(\ \ 0, +1, +2, +1, +2)$ is a legit increment schema, however the logic of ramping the cost higher than the penalty entails the notion of tradeoff loss for obtaining something that is "overpriced", and since higher scores not only are more valuable but they become exceedignly rare, it is natural to expect that cost of increments will only "increase" the more you go up.

default cutoff There is a natural cutoff setting for all schemas with the aforementioned features, called the half-volume cutoff. If one tries to get the maximum possible allowable scores (15) from D&D's standard schema for all 6 abilities, the total cost is $6 \cdot 9 = 54$ (the respective lowest cost being 0 by definition). At a first glance one could reject the half volume of 27 as a standard on account of the non-linearity of costs of acquisition; this in fact is a fallacy because, adjusting the cutoff to acount for cost non-linearities negates the desired effect of "overpricing".[2] If, however, the cutoff is adjusted to some value for any other reason, it makes it a legit choice, so this is more of a design aspect rather than a mathematicla restriction: if you mess with the cutoff be sure to do so for the right reasons.

It's easy to see that D&D's standard schema respects all of these features. Their irriducible axiomatic form is:

- baseline with zero cost of acquisition (and no cost of increment)
- strictly increasing costs of acquisition
- increasing costs of increment

[1] Note that the baseline has to be pass through by definition.
[2] This is not very precise, as the nonlinearities are still present, but have much less of an impact when treated specifically by the cutoff.

1.3. Exploring the standard schema

A schema like the ones discussed in effect defined a cost function on the domain of possible scores[3]

$$\text{cost} : \text{scores} \longrightarrow \mathbb{N}$$

where the cost of acquisition is used. In this notation, the cutoff constraint in creating a score palette takes the form

$$\sum_{\text{score}\in\text{palette}} \text{cost}(\text{score}) = \text{cutoff}$$

Such a constraint dictates a (possibly lossy) trade-off logic in creating a score: if you push one value up, you have to drom some other values down to keep the cost the same, resulting in a restricted set of allowable score palettes. Such a logic has two characteristic palettes:

a "median" that represents the most uniform score palette in the system (least score variance)

an "extreme" that represents an "edgy" score palette (most score variance)

We can build D&D's standard extreme palette by hand: start including maximum scores until the cutoff prevents adding more. In this case things are simple because 3 15s cost $3 \cdot 9 = 27$ and we 're done; the rest can only be 8, leading to $(\ 8,\ 8,\ 8, 15, 15, 15)$. This kind of extreme (half lows halfe highs) makes for an even finer requirement for a schema, that is present by default in trivial schemas (costs of increments all +1) and in carefully designed non-tricial schemas, like that of D&D.

Producing the median is not as straightforward, because the average position of the median is affected by the non-linearity of costs of acquisition. Start by increasing all ability points in the palette by one till the cutoff can no longer support it; then add the remaining points to spent the entire cutoff. We can easily see that 6 12s cost $6 \cdot 4 = 24$ so we cannot proceed to 13; we can merely make 3 of them 13, leasding to $(12, 12, 12, 13, 13, 13)$.[4]

Now it is understood that a "perfect" form for the median is that of a purely uniform score palette, but that is not always possible. However, schemas using the default cutoff will always have extreme and median of the above form (split palettes in 3 and 3 uniform subpalettes).

If one deploys all possible score palettes in D&D's standard, they will find a total of 65, the default $(\ 8, 10, 12, 13, 14, 15)$ amongst them.

2. Extending the D&D standard

An argument often heard by D&D players when choosing between rolling their ability scores or "buying" them is that while the point-buy system always yields game-balanced palettes, its score spectrum is merely 3-15. On the otherhand, rolling the best 3d6 out of 4d6 while it gives you a good change at adequate scores, there is still a considerable (and uncontrollable) risk factor that may lead to an overpowered or an underpowered player, but the score spectrum is 3-18, and this is the only reason why players may chose rolling over buying.

So ideally there would be a point-buy system that has a greater score spectrum. The official word of Wizards of the Coast is that the final word falls to the GM. However the GM naturally is afraid that if they include a new score schema they might break game balance. The purpose of this analysis is to see how can one stem new balanced schemas from an original schema that is axiomatically balanced.

2.1. Shrinkage

Before one attempts to discover how a schema naturally extends, it is enlightning to see how it naturally shrinks. The D&D standard is shown again here in Table 2.3 on the following page. It is expected that a natural shrink will happen around the median, so as a first step, one has to make sure that the new schema definitely has the same median.

[3] integer score values assumed

[4] It only falls in the middle of the schema's score range if the costs of increment are constant, which implies linear costs of acquisition.

Instead of providing a proof of why the shrinkage in Table 2.4 workswe provide some intuition. If the median is to be preserved, scores have to be removed from both ends of the spectrum. It is always safe to start by removing scores from the right (expensive) end as it dictates how many low scores you will remove from the left end. The general logic (for integer costs) is that the summed cost of increments of scores you remove from the right must match the one from the scores you remove from the left. In this example, we start by removing 15 from the right, which has cost increment +2, so we have two remove both 8 and 9 from the left as each has cost increment +1.

maximal with cutoff 51

score	4	5	6	7	8	9	10	11	12	13	14	15	16	17
cost of acqisition	0	1	2	3	4	5	6	7	8	9	11	13	15	17
cost of increment		+1	+1	+1	+1	+1	+1	+1	+1	+1	+2	+2	+2	+2

Table 2.1: maximal schema

When a "matched cost" shrinkage has been made, adjustments are required to make this a legit score schema. First we need to set the new baseline 10 cost of acquisition as the new 0. The question is do other costs of acquisition remain as is? The answer is generally no; imagine trimming the schema at a relatively high cost, it will unnaturally jump from the new zero to that high cost. There exist an intuitively natural option however: respect the cost of increments. The good thing about costs of increments is that are reference-invariant: the can apply to any cost of acquisition and generate the rest of the costs of acquisition, which makes it appropriate for applying to the new baseline 0 cost of acquisition. The results appear in Table 2.4.

increased with cutoff 39

score	6	7	8	9	10	11	12	13	14	15	16
cost of acqisition	0	1	2	3	4	5	6	7	9	11	13
cost of increment		+1	+1	+1	+1	+1	+1	+1	+2	+2	+2

Table 2.2: increased schema

Finally lets evaluate the new default cutoff; the new maximum cost of acquisition is 5 so the default cutoff is $6 \cdot 5/2 = 3 \cdot 5 = 15$. The latter $3 \cdot 5$ implies the form of the schema's extreme, which is $(10, 10, 10, 14, 14, 14)$ and matches the format implied by the original schema, i.e.: 3 lowest scores and 3 highest scores in the spectrum. The $6 \cdot 2.5 = 6 \cdot 2 + 3$ part implied the form of the schema's median; we can only increase all 6 scores in palette 2 times then spent whatever (3) is left on remaining points resulting in $(12, 12, 12, 13, 13, 13)$!

standard with cutoff 27

score	8	9	10	11	12	13	14	15
cost of acqisition	0	1	2	3	4	5	7	9
cost of increment		+1	+1	+1	+1	+1	+2	+2

Table 2.3: standard schema

Further more, it can be seen that all palettes viable by the reduced schema are in fact a subset of the standard schema's viable set! Note that the reduced spectrum is (by construction) also a subset of the standard spectrum (along with the cost of increments). So the reduced schema has

- all the features required to be a balanced system,
- the specific macro features of the schema it stems from
- its spectrum is a subset of the template's spectrum
- its viable palettes are a subset of the template's viable palettes

which in fact makes it as perfect a shrinkage possible. As an exercise, one can further try to reduce the schema in precisely the same logic[5] to yield the results of Table 2.5.

reduced with cutoff 15

score	10	11	12	13	14
cost of acqisition	0	1	2	3	5
cost of increment		+1	+1	+1	+2

Table 2.4: decreased schema

Besides being unable to reduce further this minimal schema has another interesting property: it has only one score palette, which makes $(12, 12, 12, 13, 13, 13)$ both the median and extreme! We can see by this result that maybe the 3-by-3 form for a median might eventually be preferrable to purely uniform after all. The case is different in schemas that after the proper reductions end with one value in the schema's spectrum. In such schemas, the median is expected to be uniform.

minimal with cutoff 3

score	12	13
cost of acqisition	0	1
cost of increment		+1

Table 2.5: minimal schema

[5] Matching costs of increment from left and right and keeping the same costs of increment to generate new costs of acquisition.

2.2. Extension

We can add this final feature as one needed for a score schema to really make it "perfect": a schema should eventually reduce to its median by "proper" reductions. What is more important, is that the reverse steps can be followed to stem the standard schema from the reduced one. Although the minimal schema has all too little information on how to expand, and the possibilities are numerous frankly, the reduced expansion into the standard is illuminating.

To proceed with the extension, all we have to do is respect the same rules we did for the shrinkage, however there is one complication: when adding new scores left and right the costs of increment they acquire must respect the increasing property. So any costs of increment added to the left will have at maximum the value of the minimum cost of incremeant of the previous schema (and the null value is simple shifted to the new spectrum minimum), and respectively (maximum) goes for the additions to the right. It is expected that such expansion will have all teh aforementioned good properties, for instance the standard is a genuine subset of it, its extreme is in the same form, has the same median and so on. Results are shown in Table 2.2 on the preceding page.

This schema can take one mroe extention to a "maximal" version with spectrum 4-17 shown in Table 2.1 on the previous page. The reason this is called maximal is that it is the last extension before piercing the 3-18 spectrum of rolling dice. While the full ability spectrum is 1-20, there are several modifiers that increase one of the scores by +2 points, so the theoretical schema spectrum maximummust not exceed. The "overkill" extension that can be applied to maximal leads to a 2-18 spectrum that on one hand respects the aforementioned restriction, so one could include it as a borderline case.

Summarywas mentioned

It appears that the standard schema describing how abilities points are allocated at character creation in D&D is so well defined and has mayn good properties that it enables the precise definition of possible "natural" extentions of itself. The presence of so many properties in the standard act in "advocation" of said extention by the intuition that, the more the requirements there are the moree difficult it becomes for an extension respecting them. To find extensions that survive all said restriction and inherit extra properties (like the set of viable palettes of a schema being a subset of that of its extension), we believe should act as reassurance to GMs, that they can safely use these extensions for their players, not only because they are balanced, but also because they "strictly" stem from the standard.

A. Schema compact form

It is apparent that the information of a schema can be contained in a dictionary with keys from the schema's score spectrum and values their costs of acquisition or of increment. Assuming all the good properties a balanced schema has as described in Subection 1.2 on page 1, a more compact form (zip?) can be used to generate a schema. Take the dictionary form of D&D's standard

```
{ 8: 0, 9: 1,10: 2,11: 3,12: 4,13: 5,14: 7,15: 9}
```

or its equivalent

```
{ 8: 0, 9:+1,10:+1,11:+1,12:+1,13:+1,14:+2,15:+2}
```

and compare it with the list `[ 8,13,15]`. This list, if enumerated, it has indices `[0,1,2]`. If you read the indices as costs of increments, this list tells you that score 8 is baseline because it has cost of increment 0. The next value says that up to score 13 (inclusive) costs of increment are 1. The reason is inclusive at the end is because it is non-inclusive at the start.[6] The next milestone is 15 with cost of increment 2, so all costs from the last milestone (non-inclusive) to this one should have this cost of increment. If you build a dictionary based on the process described here you will end up with the above dictionary holding the costs of increment. The former dictionary can easily stem from that by "integration" of the increments.

This means that the above compact list offers a "safe" method for defining "good" schemas from scractch. The only catch is that it makes one more assumption: the spectrum has consecutive values, a requirement however that is not unnatural and in fact is respected by almost all RPG ability schemas out there. So shall the occasion for defining a new schema arise, this representation is restrictive enough to encapsulate all good properties with minimal expressivity allowing the designer to make informed choices and have direct intuition in making a new schema.

[6] So in essence this is an attempt to save Python's convention, not break it

References

[1] Wizards RPG Team,
Dungeons & Dragons Player's Handbook,
5th edition, Wizards of the Coast (2014)